

WEEK 4: CUSTOMER ORDER AND PURCHASE ANALYSIS SYSTEM

Scenario

The Food Ordering System stores information about customers and their orders. The system needs to maintain customer order details, track the products ordered, and analyse customer purchasing patterns. These reports help the system understand order history, product demand and customer spending.

Tables Used

CUSTOMER

The CUSTOMER table stores the details of customers used for generating customer-related reports.

ORDERS

The ORDERS table stores information about customer orders, including order ID, order date, total amount and order status.

ORDER_DETAILS

The ORDER_DETAILS table stores the products, quantities and prices included in each order.

PRODUCT

The PRODUCT table stores information about the products available in the Food Ordering System.

Customer Order History Report

The Customer Order History report displays the details of customer orders.

The report includes:

- Customer Name
- Order ID
- Order Date
- Total Amount
- Status

```
SQL> SELECT FIRST_NAME AS CUSTOMER_NAME,ORDER_ID,ORDER_DATE,TOTAL_AMOUNT FROM CUSTOMER JOIN ORDERS ON CUSTOMER
.CUSTOMER_ID=ORDERS.CUSTOMER_ID;
```

CUSTOMER_NAME	ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
Aarav	0101	28-SEP-26	1500
Diya	0102	28-SEP-26	2500
Arjun	0103	28-SEP-26	1200
Meera	0104	28-SEP-26	1800

Product-wise Order Report

The Product-wise Order Report provides information about the ordering of each product.

The report displays:

- Number of times each product was ordered.
- Total quantity of each product sold.

```
SQL> SELECT P.PRODUCT_NAME,SUM(O.QUANTITY) AS TOTAL_QUANTITY FROM PRODUCT P JOIN ORDER_DETAILS O ON P.PRODUCT_ID=O.PRODUCT_ID GROUP BY P.PRODUCT_NAME;
```

PRODUCT_NAME	TOTAL_QUANTITY
Brown Rice	10
Carrots	8
Organic Apples	5
Bananas	6

Customer Purchase Analysis

Customer purchase information was analysed based on their order records.

Customers with Maximum Orders

This report identifies the customer or customers who have placed the maximum number of orders.

```
SQL> SELECT C.FIRST_NAME||' '||C.LAST_NAME AS CUSTOMER_NAME,COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME HAVING COUNT(O.ORDER_ID)=(SELECT MAX(ORDER_COUNT) FROM (SELECT COUNT(*) AS ORDER_COUNT FROM ORDERS GROUP BY CUSTOMER_ID));
```

CUSTOMER_NAME	TOTAL_ORDERS
Aarav Kumar	1
Meera Das	1
Diya Sharma	1
Arjun Raj	1

Total Spending by Each Customer

This report calculates the total amount spent by each customer based on their orders.

```
SQL> SELECT C.FIRST_NAME||' '||C.LAST_NAME AS CUSTOMER_NAME,SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME ORDER BY TOTAL_SPENDING DESC;
```

CUSTOMER_NAME	TOTAL_SPENDING
Diya Sharma	2500
Meera Das	1800
Aarav Kumar	1500
Arjun Raj	1200

Average Order Value

This report calculates the average value of orders placed by each customer.

```
SQL> SELECT C.FIRST_NAME||' '||C.LAST_NAME AS CUSTOMER_NAME,ROUND(AVG(O.TOTAL_AMOUNT),2) AS AVERAGE_ORDER_VALUE FROM  
CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME ORDER BY C.CU  
STOMER_ID;
```

CUSTOMER_NAME	AVERAGE_ORDER_VALUE
Aarav Kumar	1500
Diya Sharma	2500
Arjun Raj	1200
Meera Das	1800

Result

The required Customer Order History, Product-wise Order and Customer Purchase Analysis reports were successfully generated. The reports provide information about customer orders, product quantities, customer spending and average order values.